Pizza Sales Analysis using SQL

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Objective

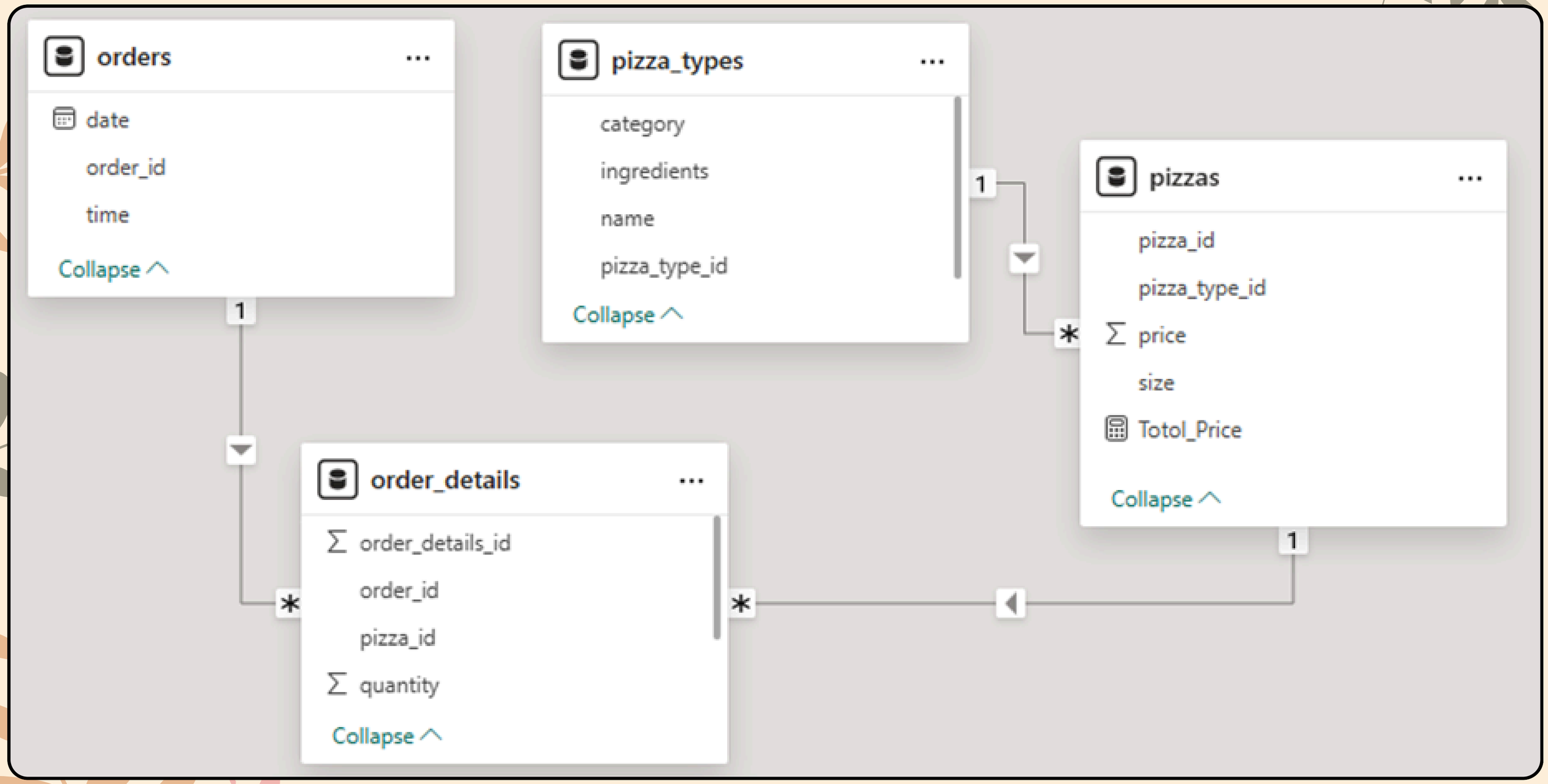
The objective of this project is to analyze pizza sales data using SQL to understand order trends, revenue patterns, and product performance. This analysis helps in identifying top-selling pizzas, peak order timings, and category-wise revenue contribution to support business decision-making.

Introduction

- This project focuses on analyzing a pizza sales dataset using SQL. The dataset contains information about customer orders, ordered pizzas, pizza prices, and pizza categories.

- By answering multiple business-driven questions, the project demonstrates the use of SQL concepts such as joins, aggregations, subqueries, and window functions to extract meaningful insights from data.

Data Model


orders
date
order_id
time
Collapse
pizza_types
category
ingredients
name
pizza_type_id
Collapse
pizzas
pizza_id
pizza_type_id
price
size
Totol_Price
Collapse
order_details
order_details_id
order_id
pizza_id
quantity
Collapse
1
1
*
*
*
1

Tables

orders: Stores order date and time information

order_id	order_date	order_time
1	2015-01-01	11:38:36
2	2015-01-01	11:57:40
3	2015-01-01	12:12:28
4	2015-01-01	12:16:31
5	2015-01-01	12:21:30
6	2015-01-01	12:29:36
7	2015-01-01	12:50:37
8	2015-01-01	12:51:37

order_details: Contains order-wise pizza quantity

order_details_id	order_id	pizza_id	quantity
1	1	hawaiian_m	1
2	2	classic_dlx_m	1
3	2	five_cheese_l	1
4	2	ital_supr_l	1
5	2	mexicana_m	1
6	2	thai_ckn_l	1
7	3	ital_supr_m	1
8	3	prsc_argla_l	1
9	4	ital_supr_m	1

pizza_id	pizza_type_id	size	price
bbq_ckn_s	bbq_ckn	S	12.75
bbq_ckn_m	bbq_ckn	M	16.75
bbq_ckn_l	bbq_ckn	L	20.75
cali_ckn_s	cali_ckn	S	12.75
cali_ckn_m	cali_ckn	M	16.75
cali_ckn_l	cali_ckn	L	20.75
ckn_alfredo_s	ckn_alfredo	S	12.75

pizzas: Contains pizza size and price details

pizza_type_id	name	category	ingredients
bbq_ckn	The Barbecue Chicken Pizza	Chicken	Barbecued Chicken, R.
cali_ckn	The California Chicken Pizza	Chicken	Chicken, Artichoke, S..
ckn_alfredo	The Chicken Alfredo Pizza	Chicken	Chicken, Red Onions, .
ckn_pesto	The Chicken Pesto Pizza	Chicken	Chicken, Tomatoes, R.
southw_ckn	The Southwest Chicken Pizza	Chicken	Chicken, Tomatoes, R.
thai_ckn	The Thai Chicken Pizza	Chicken	Chicken, Pineapple, T..
big_meat	The Big Meat Pizza	Classic	Bacon, Pepperoni, Ital.
classic_dlx	The Classic Deluxe Pizza	Classic	Pepperoni, Mushroom..
hawaiian	The Hawaiian Pizza	Classic	Sliced Ham, Pineapple.

pizza_types: Contains pizza names , categories and Ingredients

1- Retrieve the total number of orders placed

```sql
SELECT
    COUNT(DISTINCT order_id) AS total_orders
FROM
    orders;
```

total_orders
21350

Insights: A total of 21,350 orders were placed, showing strong and consistent customer demand for pizza.

2- Calculate the total revenue generated from pizza.

```sql
SELECT
    ROUND(SUM(od.quantity * p.price), 2) AS total_revenue
FROM
    order_details od
        JOIN
    pizzas p ON od.pizza_id = p.pizza_id;
```

total_revenue
817860.05

Insights: Pizza sales generated a total revenue of 817,860.05.

3- Identify the highest-priced pizza.

```sql
SELECT
    pt.name, p.price
FROM
    pizza_types pt
        JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;
```

name	price
The Greek Pizza	35.95

Insights: The Greek Pizza has the highest price at 35.95.

4- Identify the most common pizza size ordered.

```sql
SELECT
    p.size, SUM(od.quantity) AS total_quantity
FROM
    pizzas p
        JOIN
    order_details od ON p.pizza_id = od.pizza_id
GROUP BY p.size
ORDER BY total_quantity DESC
LIMIT 1;
```

size	total_quantity
L	18956

Insights: Large (L) size pizzas are the most commonly ordered, with a total quantity of 18,956, showing a strong customer preference for larger sizes.

5- List the top 5 most ordered pizza types along with their quantities.

```sql
SELECT
    pt.name, SUM(od.quantity) AS total_quantity
FROM
    order_details od
        JOIN
    pizzas p ON od.pizza_id = p.pizza_id
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_quantity DESC
LIMIT 5;
```

name	total_quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Insights: The Classic Deluxe Pizza is the most ordered pizza, followed closely by Barbecue Chicken and Hawaiian Pizza, showing these are the most popular choices among customers.

6- Join the necessary tables to find the total quantity of each pizza category ordered.

```sql
SELECT
    pt.category, SUM(od.quantity) AS total_quantity
FROM
    order_details od
        JOIN
    pizzas p ON od.pizza_id = p.pizza_id
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_quantity DESC
;
```

category	total_quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Insights: The Classic category has the highest total orders, followed by Supreme, indicating these categories are the most popular among customers.

7- Determine the distribution of orders by hour of the day.

```sql
SELECT
    HOUR(order_time) AS order_hour,
    COUNT(order_id) AS orders
FROM
    orders
GROUP BY order_hour
ORDER BY orders DESC;
```

order_hour	Orders
12	2520
13	2455
18	2399
17	2336
19	2009
16	1920
20	1642

Insights: The highest number of orders are placed at 12 PM and 1 PM, showing peak demand during lunch hours.

8- Join relevant tables to find the category-wise distribution of pizzas.

```sql
SELECT
    pt.category, COUNT(p.pizza_id) AS total_pizzas
FROM
    pizzas p
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_pizzas DESC;
```

category	total_pizzas
Veggie	27
Classic	26
Supreme	25
Chicken	18

Insights: The Veggie category has the highest number of pizza varieties, followed closely by Classic.

9- Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
SELECT
    ROUND(AVG(quantity), 2) AS avg_pizza_ordered_per_day
FROM
    (SELECT
        o.order_date, SUM(od.quantity) AS quantity
    FROM
        orders o
    JOIN order_details od ON o.order_id = od.order_id
    GROUP BY o.order_date) AS order_quantity;
```

avg_pizza_ordered_per_day
138.47

Insights: On average, about 138 pizzas are ordered per day, showing consistent daily demand.

10- Determine the top 3 most ordered pizza types based on revenue.

```sql
SELECT
   pt.name, SUM(od.quantity * p.price) AS revenue
FROM
    order_details od
        JOIN
    pizzas p ON od.pizza_id = p.pizza_id
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Insights: The Thai Chicken Pizza generates the highest revenue, followed by Barbecue Chicken and California Chicken, making them the top revenue-contributing pizzas.

11- Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
    pt.category,
    CONCAT(ROUND(SUM(od.quantity * p.price) / (SELECT
                            SUM(od2.quantity * p2.price)
                        FROM
                            order_details od2
                                JOIN
                            pizzas p2 ON od2.pizza_id = p2.pizza_id) * 100,
                    2),
            ' %') AS revenue
FROM
    order_details od
        JOIN
    pizzas p ON od.pizza_id = p.pizza_id
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY revenue DESC;
```

category	revenue
Classic	26.91 %
Supreme	25.46 %
Chicken	23.96 %
Veggie	23.68 %

Insights: Classic pizzas contribute the highest percentage of total revenue.

12- Analyze the cumulative revenue generated over time.

```sql
SELECT
    order_date,
    SUM(daily_revenue) OVER (ORDER BY order_date) AS cumulative_revenue
FROM (
    SELECT
        o.order_date,
        SUM(od.quantity * p.price) AS daily_revenue
    FROM orders o
    JOIN order_details od
        ON o.order_id = od.order_id
    JOIN pizzas p
        ON od.pizza_id = p.pizza_id
    GROUP BY o.order_date
) AS daily_sales
ORDER BY order_date;
```

order_date	cumulative_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4

Insights: Each day's revenue is added to the previous total, so the revenue keeps increasing over time.

13- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```sql
SELECT  category, name, revenue
  FROM ( SELECT category, name, revenue,
         DENSE_RANK() OVER (
             PARTITION BY category
             ORDER BY revenue DESC
         ) AS rank_in_category
  FROM (SELECT pt.category,   pt.name,
             SUM(od.quantity * p.price) AS revenue
         FROM order_details od
         JOIN pizzas p   ON od.pizza_id = p.pizza_id
         JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
         GROUP BY  pt.category, pt.name
     ) AS category_revenue
) AS ranked_pizzas
WHERE rank_in_category <= 3
ORDER BY category, revenue DESC;
```

category	name	revenue
Chicken	The Thai Chicken Pizza	43434.25
Chicken	The Barbecue Chicken Pizza	42768
Chicken	The California Chicken Pizza	41409.5
Classic	The Classic Deluxe Pizza	38180.5
Classic	The Hawaiian Pizza	32273.25
Classic	The Pepperoni Pizza	30161.75
Supreme	The Spicy Italian Pizza	34831.25
Supreme	The Italian Supreme Pizza	33476.75
Supreme	The Sicilian Pizza	30940.5

Insights: Each day's revenue is added to the previous total, so the revenue keeps increasing over time.

Thank You